

We built a reconciliation data platform — then used it to test how AI should review code

A real client needed reconciliation data made legible. We shipped a production platform for the **Indigenomics Institute** — and its own **30 merged pull requests** became the industrial dataset for a pre-registered study of multi-agent AI code review.

Research companion: “Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study”

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li

Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

RESEARCH CONTRIBUTION

More AI reviewers don't help — **decorrelation does**. Agreement predicts truth only when the reviewers are independent.

89%

3 independent families agree → matches ground truth

54%

one model, three runs → mostly noise

01 THE PROJECT — RAP DATA PORTAL

PROBLEM

Companies publish reconciliation commitments. Legal precedents sit scattered across court records. There is no searchable, verifiable, unified view — so nobody can track progress.

OUR SOLUTION

▪ Searchable RAP commitments ▪ Legal precedent database ▪ AI extraction from PDFs ▪ Human review workflow ▪ AWS serverless hosting
▪ Authentication & roles ▪ Progress tracking ▪ Coverage analytics

WHO USES IT

▪ Indigenomics Institute ▪ Indigenous organizations ▪ Corporate partners ▪ Researchers & legal advisors

Indigenomics · RAP Index

INDIGENOMICS RAP Index Organizations Suppliers Coverage analysis Verification Alignment Extract Notifications

Table Explore

The RAP Index commitments by sector, size & type
Reconciliation commitments across the network, and how they progress over time. Distinct from Spend Coverage, which tracks confirmed dollars.

Key takeaways
Plain-language highlights, generated automatically from the data below. Start here.
• 106 commitments across 100 organizations, averaging 63% progress. 0% already confirmed.
• Procurement is the most common commitment type (36), while Employment lags on delivery at 53% average progress.
• Of 29 claimed outcomes, 0% are supplier confirmed, 29 remain self-reported and unverified.
• Needs attention: 31 are past their target year without confirmation, and 43 due in 2026 are behind pace.
Generated from the data below - reflects the current filter.

At a glance
The headlines totals for the current view: commitments, organizations, average progress, and the confirmed share.
106 commitments 100 organizations 63% average progress 0% confirmed

Indigenomics · Legal Cases

Legal Cases Cases Find similar Activation Legal info Monitoring Methodology

Unofficial reproductions of public court decisions - not legal advice - every claim links to its source

Legal cases — economic justice
Canada's Indigenous economic justice case law, searchable and citation-anchored.
Describe your situation to find similar cases →
View as: Indigenous government Legal advisor Corporate / advisory
Precedents affirming your community's economic rights and self-determination. *The same public record, re-rendered for your context — anyone can switch, noting its history.
Search citation, case name, or full text...
Tier: auto All themes All courts All outcomes All nations from yr to yr clear
561 results · showing 1-10 · browse · start case

Legal Cases
• 561 economic-justice decisions
• citation-anchored, role-based views

RAP Index
• 106 commitments · 100 organizations
• 63% average progress tracked

ARCHITECTURE — SERVERLESS ON AWS

02 HOW IT BECAME RESEARCH

Client problem

reconciliation data scattered & unverifiable

We built the portal

production platform, shipped on AWS

179 pull requests

30 of them ours — real, merged, AI-reviewed

Benchmark dataset

E1 Qodo · E2 SWE-PRBench · E3 our portal — 2,415 reviews

Four AI review architectures

each rung adds exactly one capability

03 WHAT WE FOUND · E1 · Q000, 99 PRS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling (p=0.73); peer debate adds nothing over a manager (p=0.12).

Extra agents buy recall at a price — up to 5x the cost per confirmed finding.

Architecture	Recall
Agentless (1 call)	0.34
Generalists-3 (3 calls)	0.45
Hierarchical (3 calls)	0.50
Consensus (9 calls)	1.76

5x more expensive per confirmed finding than single-pass review

Bounded. Recall plateaus at ≈0.83; on our own unlabeled code, LLM judges disagree (κ=0.03).

WHERE THE SIGNAL LIVES

Agreement recovers **universal functional bugs** far better than project-specific conventions — which is why the ceiling exists.

43%
functional-bug recall at 3 independent families agreeing

18%
rule-violation recall at the same depth

Pair decorrelated review with a deterministic linker.

PROJECT IMPACT

- Production platform **delivered** to the Indigenomics Institute
- Reconciliation commitments made **searchable and verifiable**
- Indigenous legal precedent **explorer**, citation-anchored

RESEARCH IMPACT

- First **compute-matched** comparison of review architectures
- **Independence beats more agents** — decorrelate your verifiers
- Guidance for anyone deploying AI code review in production

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Pre-registered on OSF · registration 224vj (embargoed) · github.com/SharonHuang77/rap-review-research